



Installation Manual VG

Document Number: SMI-0248

Revision: 3.0

Status: Approved

Date: 3 March 2009

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Revision History

Revision	Date	Author	Type (Major/Minor)	Summary of Changes
1.0	20 NOV 2007	Dylan Laister	Major	Initial Release - Created instructions during test install
2.0	23 JUN 2008	Chris Luginbuhl	Minor	Added service tools, figures
3.0	2 MAR 2009	Dylan Laister	Minor	Minor edits for clarity, GE V&V, new figures for engineering changes.

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1.0 Purpose

This Installation Manual is designed to provide detailed instructions to an SMI Service Technician for installation and testing of the Vanguard VG System including the stretcher/patient support and RF Coils (mechanical and electrical parts).

If any non-conformances or problems are discovered during installation, it is imperative that these are recorded, reported and ultimately entered into the SMI Problem Management Database.

2.0 Service Tools Required

Item	Part Number	Qty	Description
1	SMI-0294	1	Service Toolkit

3.0 Definitions and Abbreviations

Throughout this document the following terms and abbreviations will be used:

UBS	– Upper Body Support
LBS	– Lower Body Support
FSA	– Front Support Assembly
RSA	– Rear Support Assembly
Head end	– Patient Head end of the Vanguard
Foot end	– Patient Foot end of the Vanguard

Stretcher	– Bottom portion of the Vanguard
Patient Support	– Top fibreglass portion of the Vanguard
Left/Right	– Directions relative to the patient's position

4.0 Precautions to be taken

Before entering the MRI Room with the Vanguard system the following activities must be performed:

- Thoroughly inspect the system to ensure loose ferrous/magnetic items are not present anywhere on the system. Loose objects can become airborne and/or travel at high velocity in the presence of the scanner, potentially causing harm to the Patient, Operator or equipment.
- Do not attempt to make any changes or adjustments to the GE Signa scanner or GE Signa stretcher. The Vanguard is designed to be adapted to the current state of the GE Signa scanner operating in a normal mode. If any damage to the GE Signa scanner is observed report to the customer.
- Do not touch metal pins of any connectors. Leave the protective packaging on all connectors until install is complete to prevent electrostatic discharge damage.
- Patient data is confidential – be sure not to examine, copy or photograph any printed or electronic matter containing patient names, personal details, medical images or other confidential data to ensure compliance with the Health Insurance Portability and Accountability Act (HIPPA).

5.0 Preparing for the Service Call

Prior to performing the Service Call the following are required:

- Installation should be booked a minimum of two weeks before application training is scheduled to start. Follow up 3 days before the installation date to remind the customer and confirm that the scanner time is booked off.
- A copy of the P.O. or work order for the unit to be installed.
- Contact information for the site (usually the charge tech or administrator) and back-up contact person.
- Determine if any nearby sites can be visited during the same trip to perform preventative maintenance.
- Bring a complete service kit, camera (if possible), notebook, service report cards and personal business cards.
- When international travel is required, a signed letter from the service manager or other manager explaining the purpose of the trip, the destination, dates and the employment status of the service technician.
- ID badge, to be kept on person at all times, and passport/visa (as needed).
- Hotel info, customer site address, directions to/from site, airport and hotel.
- If possible, contact the site specific GE Service Personnel to inform them of the pending service call and request information on any recent problems with or service to the scanner (SMI service calls can be generated from recent service to the scanner involving adjustment of mechanical parts or software upgrades and patches).

- Ensure that only Sentinelle-authorized service personnel make adjustments or repairs to the system.
- Review Sentinelle service logs for the site for previous service reports and service pending.

6.0 Performing and Recording the Service Call

Record the results of all activities performed including:

- Record any non-conformances that were discovered during unpackaging and installation or service and any remedial action taken.
- Leave a completed service card and inform relevant staff members who are present that service is complete and indicate state of functionality. If the system fails any of the functionality tests in the installation procedure, it must be taken out of service.
- Unless prior arrangements have been made with the Applications Support Specialists, the system will normally be “Out of Service – Pending Applications training” immediately following installation. *Site staff should be informed of this.*
- A SMI or affiliate service log must be filled out to officially record the installation.

7.0 Installation Procedure

7.1 Initial Alignment

The initial alignment sets up the Vanguard for install and assesses the Vanguard for height adjustment. No attempt to dock the Vanguard should be made at this time. The patient support is heavy and fragile. Exercise caution when moving the patient support and get another person to help lifting if possible.

1. Undock GE Signa table and move out of work area.
2. Remove the patient support from the stretcher and place in a secure location by:
 - Grasping the patient support in the middle, lifting it clear of the stretcher and placing it on any horizontal surface, such as a counter top or the Signa table, taking extra care not to damage any of the patient support hardware or the Signa table.
 - OR
 - Pressing the dock pedal and tilting the table slightly clockwise to clear the doghouse interlock, while lifting the head end and pulling the patient support past the UBS. Place bottom half of the patient support on the UBS handle between the last two sets of wheels. By rotating the patient support on the UBS handle, it may then be placed head end first on the floor and carefully leaned against a wall. Ensure that the patient support is not scratched, scuffed or damaged while doing so.
 - Using packing material or linens to protect the patient support, Signa table or other items it may be in contact with.

3. Remove the drawers from the stretcher by depressing the two plastic latches on either side, downwards on the left and upwards on the right, and pulling the drawers out of the rear support assembly (Fig 7.1.1). Place them away from the work area. The drawer sliders have a light oil coating which may get on hands, and if so, should be washed off to avoid fingerprints on the system.



Fig 7.1.1

4. Check to ensure the doghouse sensor actuator in the middle of the bumper is screwed in as far as possible turning it in a clockwise direction by hand (Fig 7.1.2).



Fig 7.1.2

5. Use your finger to depress the MRI's docking pin (circled in Fig 7.1.3) until a slight click is heard and hold it while pressing the scanner control button to advance the

carriage or “doghouse” into the bore of the scanner (away from you) until it stops.

6. Align the Vanguard to the scanner carefully by rolling it up to the centre of the scanner, ensuring that the bumper does not contact the fixed static pin protruding from the dock. The static pin should be aligned with the corresponding clearance pocket in the bumper (Fig 7.1.3). Measure the height difference of the stretcher and the MRI's bridge (bottom surface inside the bore) with an MRI compatible ruler and straightedge or other available means.

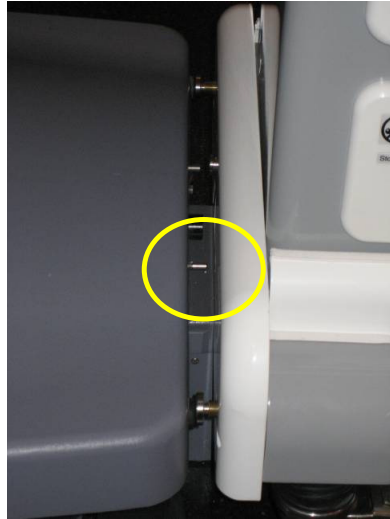


Fig 7.1.3

7. If the height difference between the stretcher and the MRI's bridge is greater than 1/2", perform coarse height adjustment (Section 7.2), followed by fine height adjustment (Section 7.3). Coarse adjustment can only be performed in increments of 1/2". If the height difference is less than 1/2", skip the coarse height adjustment section and proceed to the fine height adjustment section.

7.2 Coarse Height Adjustment

Coarse height adjustment is performed when the height difference between the MRI's bridge and the Vanguard LBS is 1/2" or greater.

1. Loosen the cable which actuates the no-undock hook by loosening the round head screw which clamps the cable to the hook with a 5/32" Allen key (Fig 7.2.1).

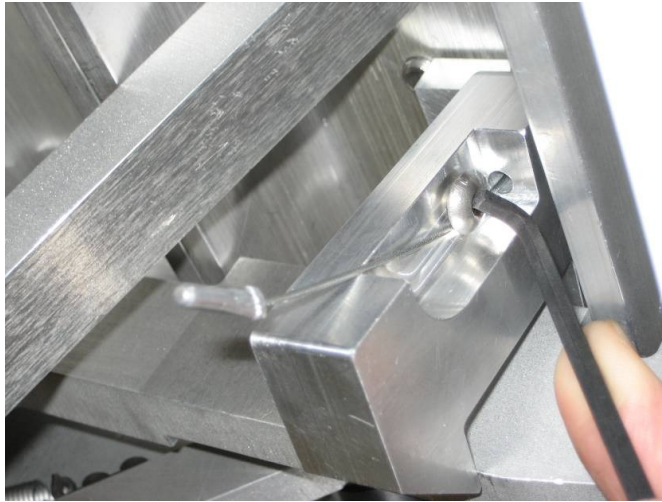


Fig 7.2.1

2. Loosen the cable which actuates the latch tab by loosening with a 5/32" Allen key the socket head cap screw which fastens the cable clamp to the swing assembly (Fig 7.2.2).

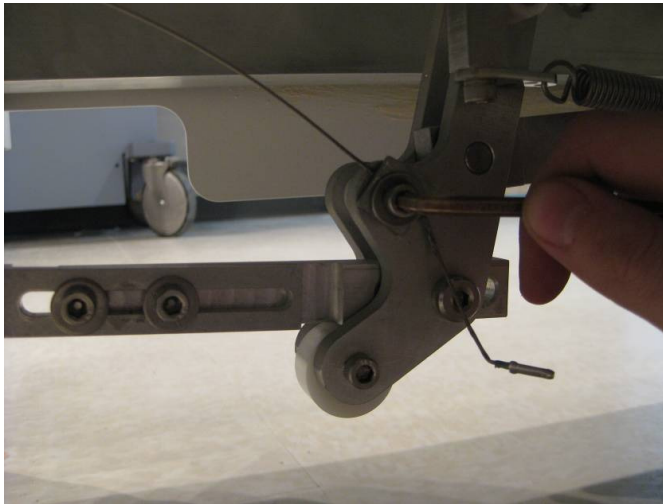


Fig 7.2.2

3. Using a 1/8" Allen Key remove the LBS skirt by unscrewing the two external mounting round head screws and loosening the third screw. Slide the skirt down to the bottom of the RSA and remove it by bending it slightly (Fig 7.2.3). Be careful not to scratch any of the paint.



Fig 7.2.3

4. Remove the two half skirts from the RSA of the stretcher by unscrewing the four round head screws on each side using a 5/32" Allen key (Fig 7.2.4).



Fig 7.2.4

5. Open the battery compartment access cover on the FSA by unscrewing the thumb fastener (Fig 7.2.5) and then pivoting the assembly into its open position (Fig 7.2.6).



Fig 7.2.5



Fig 7.2.6

6. Take note of the position of the 4 bolts in the series of vertical holes inside the FSA and RSA. A marker may be used if necessary (Fig 7.2.7).



Fig 7.2.7

7. Inside the RSA, completely loosen four nuts and lock washers using a 1/2" wrench. From the outside of the RSA remove two of the four carriage bolts which are nearest to the bumper while supporting the LBS (Fig 7.2.8).
8. ***Caution: do not remove all four of the carriage bolts at the same time as the LBS will be unsupported and could cause injury or damage. The following steps (9, 10) detail, and will only permit, height adjustment one increment at a time. In the rare case when more than one increment is required, repeat each step more than once.**
9. Adjust the LBS height by tilting the LBS up or down while positioning one of the carriage bolts in its mounting hole so it will slide in as the adjustment holes lines up. Insert the other bolt ensuring the square base of the shank is correctly oriented to the square hole in the RSA. Repeat the process on the other side of the LBS and fasten with the four sets of nuts and lock washers.



Fig 7.2.8

10. Adjust the UBS while facing the open battery access door in the side of the FSA. Remove the 2 closest and one of the farther bolts while supporting the UBS. Once the bolts are removed this will cause the UBS to tilt down from the point where the carriage bolt remains. Tilt the UBS up or down one adjustment increment in both axes positioning the opposite carriage bolt in its mounting hole so it will slide in as the adjustment holes line up. Remove the bolt which remained at the initial height and level the UBS in the same manner. Finally, push the remaining carriage bolts into place when the correct height is set, re-install the 4 sets of nuts and lock washers.
11. Upon completion of coarse height adjustment, verify the height adjustment and then replace the skirts on the UBS and LBS.

7.3 Fine Height Adjustment

Fine height adjustment is performed when height adjustment of less than 1/2" is required or after coarse height adjustment has been performed. This adjustment is essentially the "ride height" determined from adjusting the casters, which are mounted to the head of a large bolt which is then screwed into a plate in the stretcher frame.

1. Align the stretcher to the scanner carefully, ensuring that the bumper does not contact the fixed static pin or the doghouse switch on the dock, and take note of the height difference.

2. Remove the wheel covers from the stretcher by loosening the two bottom-mounted round head screws several turns using a 5/32" Allen key and sliding the covers off (Fig 7.3.1).

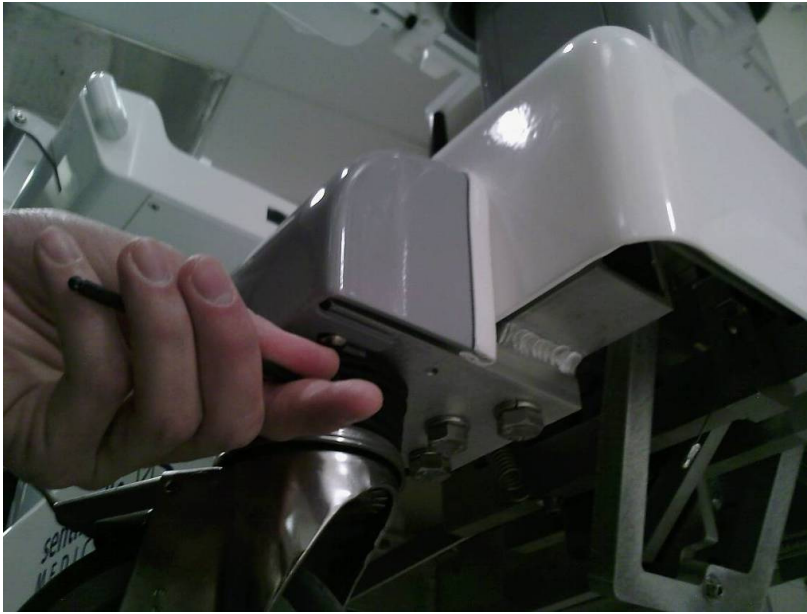


Fig 7.3.1

3. Loosen each of the large nuts so they are flush with the end of the bolt, using the 1.5" MRI compatible wrench (Fig 7.3.2).



Fig 7.3.2

4. Compress the black rubber shroud, place the 1.5" MRI compatible wrench on the exposed height adjustment bolt head and turn clockwise (as seen from above) to raise the stretcher, or counter clockwise to lower the stretcher. Ensure that all four hex bolts are initially adjusted by the same number of turns (**Note that a 360 degree turn = 1/8" of height adjustment**). This can also be accomplished by locking the caster, raising the edge of the stretcher and rotating the caster with one foot.
5. Sighting along the length of the flat surface of the stretcher and the scanner's bridge, adjust the wheels at the head or foot end to correct approximately for any pitch or roll angle.
6. Next, use the flat straightedge to measure the co-planarity of the stretcher surface and scanner's bridge. First, lay the straightedge on the scanner bridge, and attempt to slide the tool out over the stretcher. If it clears the stretcher, measure the gap. Repeat this process on both sides (left and right) of the bridge. If the tool will not slide over the stretcher, then lie the straightedge on the stretcher's Lower Body Support (LBS) and attempt to slide it out over the scanner's bridge. Ensure that the tool slides freely past the ramped section of the bridge, and continues to slide freely at least 6" into the scanner. Measure the gap under the tool in both left and right wheel tracks.
7. Adjust the caster height by the measured amount from the higher surface. The lower body support must be level or no more than 1mm (3/64") lower than the scanner's bridge, disregarding the sloped region of the bridge (Fig 7.3.3).

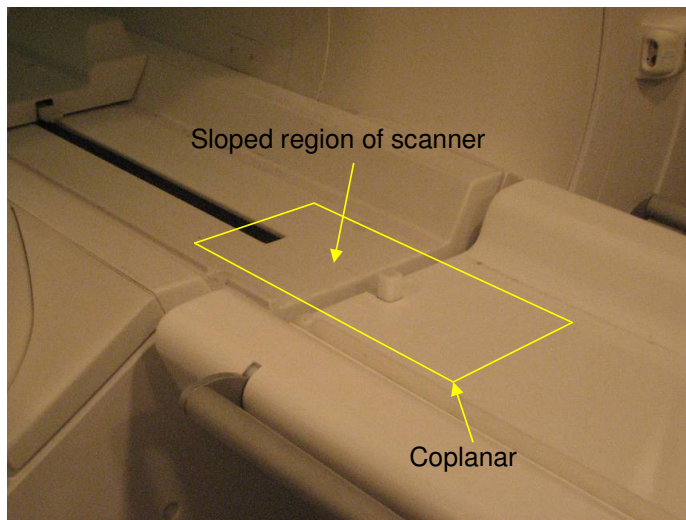


Fig 7.3.3

8. Once fine height adjustment has been performed, re-check the pitch of the stretcher relative to the scanner bridge by sighting along the stretcher into the bore and using the straight edge. Perform fine adjustment as required while continually checking for co planarity.

***Note adjusting the pitch with the head end casters will affect the height of the edge of the LBS foot end relative to the scanner, as the pitch angle will rotate about the casters at the foot end of the stretcher. The fine height adjustment must be verified again within 1 mm, and re-adjusted if necessary. Fine height adjustment is extremely critical to smooth and reliable operation. It is always worthwhile taking a little extra time to get it perfect.**

9. Scanner room floors are rarely flat. Verify that all the stretcher wheels are in contact with the floor when the stretcher is in the docked position. This can be checked by trying to swivel the casters with the stretcher's weight on them to determine if it is evenly distributed. Adjust any non-contacting casters to ensure all four wheels make contact. Sometimes this will result in "wobble" of the stretcher when not in the dock position to the scanner, which is unavoidable.
10. Tighten each of the large nuts very tightly using the 1.5" non-magnetic wrench. Ensure the fine height adjustment bolts do not spin when the nuts are tightened. It is necessary to brace against the stretcher and use both hands for very high torque approximately 1/8 turn past tight.
11. Replace the wheel covers on the stretcher and tighten the round head screws using a 5/32" Allen key, ensuring the covers fit tightly to the shroud.

7.4 Docking Hook Height Adjustment

Docking hook height is set from the factory but is affected when fine height adjustment of the stretcher is made.

1. Place the stretcher close to the dock so the stretcher docking hook enters the dock.
2. Loosening the socket head cap screw with a 3/16" Allen key and adjust the position of the round eccentrically mounted hook stop so the docking hook is at least 1/4" above the scanner's bronze dock hook and is not contacting the inside top surface of the dock. (Fig 7.4.1).



Fig 7.4.1

7.5 Bumper Adjustment

The guide rails on the top of the stretcher and on the scanner bridge must be collinear in order for the patient support to advance into the scanner. Simultaneously, the stretcher bumper must be centered vertically and horizontally with respect to the Dock to permit function of the Vanguard.

1. Loosen the two 1/4"-20NC bolts on the back face of the bumper which lock the LR adjustment bumpers in place with a 3/16" Allen key (Fig 7.5.0).



Fig 7.5.0

2. Loosen the two left-right fine adjustment bumpers which are located in slots on the bottom of the bumper with a 1/4" Allen key and slide them into the bumper to their fullest extent (Fig 7.5.1).

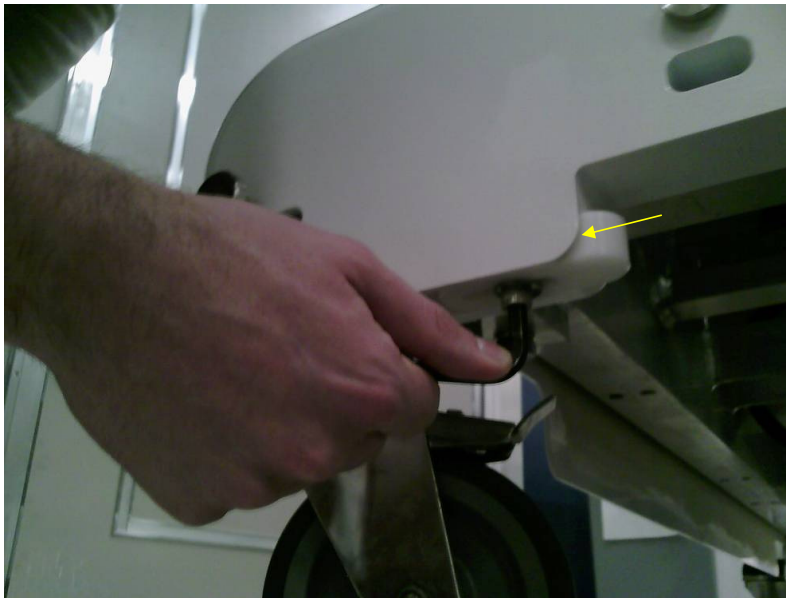


Fig 7.5.1

3. Move the stretcher to the dock position against the scanner and then back so the bumper is roughly 2" from contacting the scanner dock to permit an Allen wrench to be used. Kneel at the head end of the stretcher and sight along the guide rails and position the stretcher so the guide rails are collinear (Fig 7.5.2). Lock all of the casters ensuring the stretcher position is not affected while doing so.

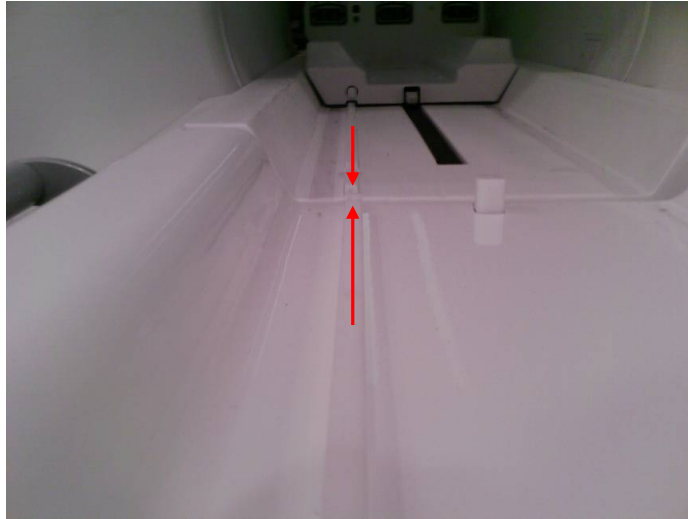


Fig 7.5.2

4. The bumper spring plungers of the stretcher bumper must now be centered to the corresponding rubber features of the dock. Take note of the amount that it must be moved in order to centre it (Fig 7.5.3).

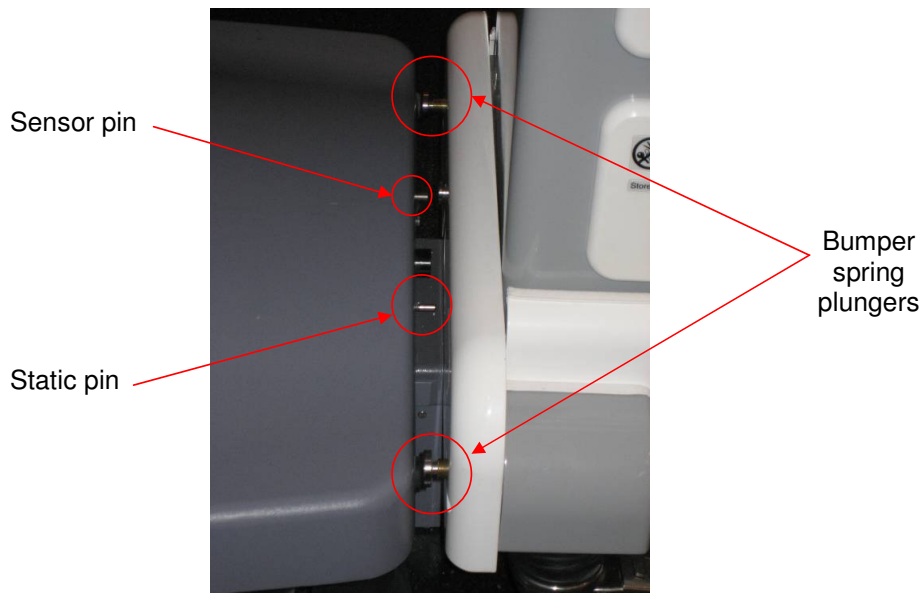


Fig 7.5.3

5. View the bumper alignment from the top and then from the side. Loosen the four socket head cap screws with a 1/4" Allen key (Fig 7.5.4). Adjust the position of the bumper plate vertically and horizontally as required and tighten the screws. Push the stretcher against the dock and check the alignment of the bumper features. The spring bumpers should be concentric with the black rubber bumpers on the dock. If not, unlock casters and move stretcher back and reposition bumper as required.



Fig 7.5.4

6. After the bumper is centered, firmly tighten the four socket head cap screws to secure the bumper plate to the stretcher using a 1/4" Allen key, supporting it to ensure it doesn't shift while being tightened.

7.6 Docking Force Adjustment

Docking force is an important aspect of the function of the Vanguard. Setting the docking force requires incremental adjustments of the docking hook until adequate docking force is achieved. If the docking force is too high it will result in premature wear to the Vanguard docking mechanism. Too little docking force will not be sufficient to keep the stretcher immobilized during use and can cause it to undock unexpectedly.

1. Loosen the two socket head cap screws on the docking hook assembly (Fig 7.6.1) using a 3/16" Allen key and adjust the hook by sliding it to the shortest position. Leave the two bolts loose for now.

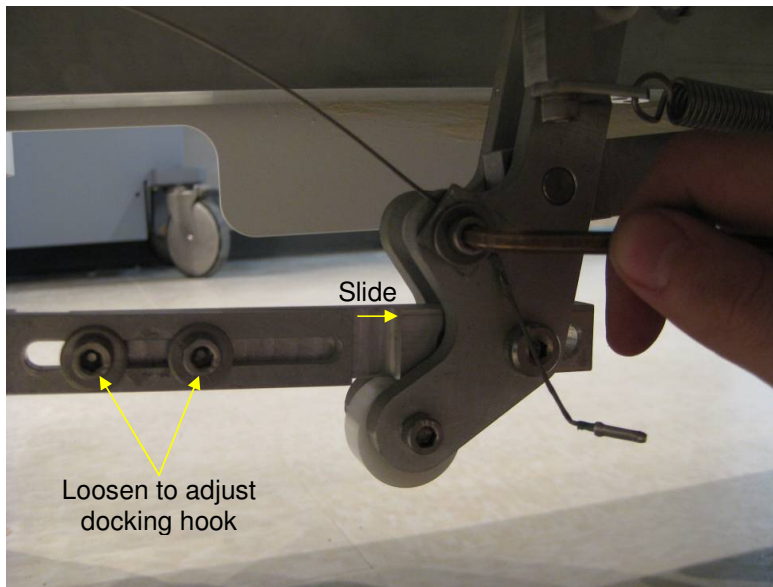


Fig 7.6.1

2. While applying constant force toward the scanner from the UBS handle of the stretcher to compress the bumper spring plungers, depress the docking pedal and then release it slowly to prevent any jarring motion to the docking hook, then release force from the UBS handle. This sets the hook to the point where it contacts the bronze hook in the dock while the dock pedal is completely depressed.
3. Adjust the hook by sliding it 1/16" away from the dock, and tighten one of the two socket head cap screws using a 3/16" Allen key.
4. Take note of the docking pedal travel with the stretcher away from the scanner. Place the stretcher against the dock and begin depressing the docking pedal slowly and take note of the peak force required to dock the Vanguard, which will occur in the last quarter of pedal travel. If the force required to dock will require more than the weight the foot and leg being placed on the pedal, stop at once and do not press any further—loosen and extend the docking hook approximately 1/32" and retighten.

***Note: the docking pedal force must be between 18 and 22 lbs., which is provided by the weight of an average foot and leg on the pedal and only very light muscular force.**

5. Test the docking force adjustment while the stretcher is docked by attempting to push the stretcher sharply towards the scanner with considerable force. If it comes undocked increase the force slightly by loosening the bolts, retracting the hook a further 1/32" and repeat step 4.
6. When the stretcher is docked, verify that the bumper spring plungers remain fully compressed even when pushing on the sides of the stretcher. If the stretcher comes undocked or the bumper spring plungers are free to telescope slightly, adjust the docking hook tighter in 1/32" increments while repeating steps 4-6.
7. When adequate docking force is achieved, firmly tighten the two socket head cap screws on the docking hook assembly using a 3/16" Allen key.

7.7 Bumper spring plungers and Straightness (Yaw) Adjustment

The bumper springs can be adjusted to correct for any slight misalignment between the guide rails of the stretcher and scanner

1. Align the guide rails and gently dock the stretcher to the scanner. While doing so, gauge by eye or use a long straight edge to assess the straightness or "yaw angle" of the stretcher and scanner guide rails.
2. Undock the stretcher and ensure the guide rails are collinear (Fig 7.7.1) by turning the threaded brass body of one bumper spring plunger. For every turn or fractional turn, apply the *opposite* number of turns to the other bumper spring plunger (Fig 7.7.2). Applying the opposite adjustment on each side ensures no increase or decrease of docking force. Dock and undock the stretcher to check the effectiveness of the alignment adjustment.
3. Once the stretcher is as straight as possible, double check that the dock force is not too high, and that the bumper spring plungers stay completely bottomed out even when the stretcher is pushed towards the scanner or side-loaded (see sec. 7.6)

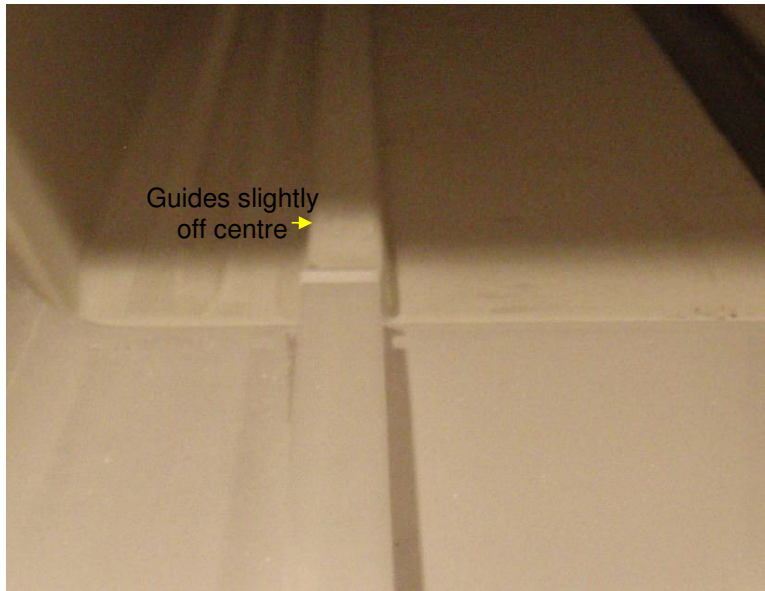


Fig 7.7.1



Fig 7.7.2

4. Now, with the stretcher docked, adjust the height of the center plunger. It is possible to overextend the bumper's center plunger and thereby damage the scanner's dock switch. It is also possible to depress the switch too little, which leads to erratic operation. The ideal is to depress the switch halfway between where it actuates (clicks) and where it bottoms out. Extend the center plunger by unscrewing it by hand until it just depresses the sensor pin on the dock - you will hear a faint click - then adjust it out a further 1/8" to 3/16"

7.8 Final Centering Adjustment

Centering of the Vanguard is an adjustment which requires attention to detail and must be executed with precision. This adjustment will ensure that the patient support will transit to the scanner's bridge smoothly.

1. Dock the stretcher by slowly depressing the docking pedal while carefully aligning the guide rails of the stretcher and the Scanner's bridge using a straightedge. It may take more than one attempt to get the guide rails perfectly aligned
2. Slide the two left-right fine adjustment bumpers (located on the bottom edge of the bumper plate, (fig 7.5.1) so they firmly contact the scanner's dock. Fasten them firmly in place by tightening the socket head cap screws with a 1/4" Allen key while pressing them against the dock with your other hand.
3. Undock, reposition the stretcher and dock the stretcher again to ensure that it is centered. Re-adjust the left-right bumpers if required and check again. When the stretcher is pressed up against the scanner but undocked, there should be no more than 1/16" of L-R play possible by applying lateral force to the stretcher near the foot end. Once the stretcher is centered with bumpers firmly fastened, undock the stretcher and firmly tighten the two socket head cap screws on the back side of the bumper plate with a 3/16" Allen key (fig 7.5.0).

***Note – subtle left-right misalignment (Fig 7.7.1) is sufficient to cause serious problems and will prevent long-term smooth and reliable operation.**

7.9 Stretcher to Scanner Gap Adjustment

The gap adjustment is required to eliminate a possible pinch point between the Vanguard LBS and the MRI's bridge. This adjustment involves moving the LBS in the head foot direction, and later (Section 7.11) moving the UBS using the patient support as a gauge.

1. Dock the stretcher to the scanner and measure the gap between the stretcher LBS and the scanner with a MRI compatible ruler. This gap must be 1" (+ 0.25/ -0) when the system is docked (Fig 7.9.1).



Fig 7.9.1

2. If adjustment is required, move the LBS by loosening with a 9/16" wrench or ratchet the four hex head bolts which are located inside the LBS. Access these bolts by reaching through the drawer openings in the RSA and upward to locate the bolt heads on the top of the RSA plate on which the LBS is mounted (Fig 7.9.2). Set the gap to 1" (Fig 7.9.1). When complete, tighten the four bolts firmly to fasten the LBS in place.



Fig 7.9.2

7.10 Latch Tab Height Adjustment

The latch tab keeps the patient support in place when the Vanguard is not docked and like the GE Signa table, unlatches the doghouse from the patient support when the Vanguard is undocked.

1. Dock the Vanguard to the scanner.
2. Using pliers, pull the cable far enough to ensure the latch tab is flush or below the LBS surface (Fig 7.10.2). Clamp the cable in place by tightening the cable clamp screw with a 5/32" Allen key (Fig 7.10.2).
3. Undock and dock the table and take note of the latch tab height. Undock the stretcher loosen the cable clamp and take up just enough cable to bring the latch tab flush when docked., Ensure that the travel of the latch tab

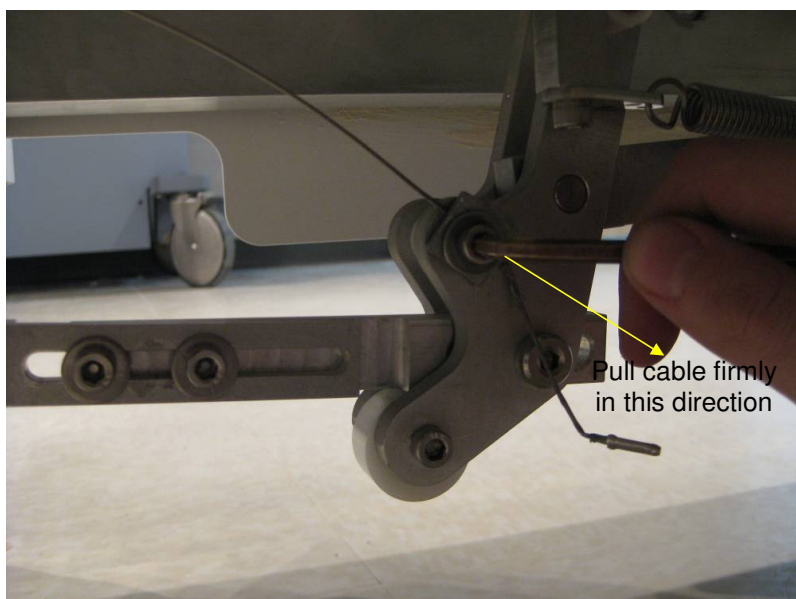


Fig 7.10.2



Fig 7.10.3

7.11 Head End Position Adjustment

The Upper Body Support (UBS) can be adjusted in the Head-Foot direction, as well as the Left-Right direction. The Left Right direction helps ensure the bridge latches smoothly, and the Head-Foot direction determines the travel stop position of the patient support.

1. Open the bridge. Prepare to reposition the UBS by loosening the four hex head bolts on the underside of the UBS using a 9/16" socket or wrench (Fig 7.11.1). Pull the UBS handle to slide the UBS all the way to the head end.
2. Replace the patient support on the stretcher.

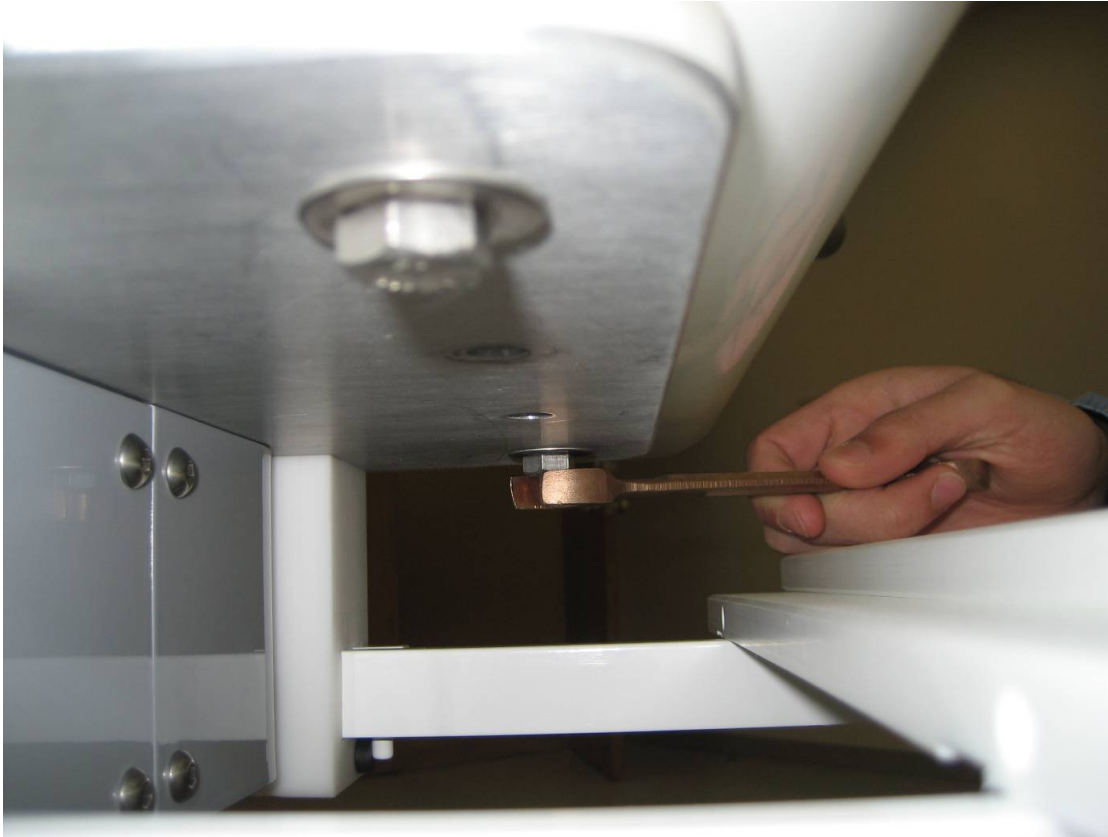


Fig 7.11.1

3. Push the UBS gently toward the foot end until the travel stop (fig 7.11.4) on the patient left side of the UBS bottoms out against the patient support.
4. Close the stretcher's bridge most of the way so that its spring plungers are just contacting the LBS (Fig 7.11.2).. Adjust the UBS so that the bridge's nose doesn't rub the side of the square hole in the LBS.

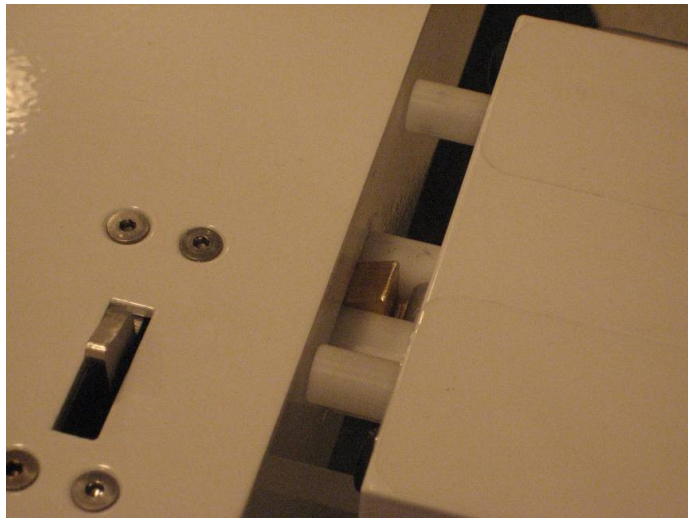


Fig 7.11.2

5. Ensure there is a gap between each side of the latch tab and the latch fixture (Fig 7.11.3) by firmly pushing the patient support towards the head end of the stretcher with a 1/32" spacer in the gap. The latch tab must not rub any part of the latch fixture when the patient support is at its home position on the stretcher.



Fig 7.11.3

6. Ensure the UBS is centered left/right relative to the tabletop by looking at the travel stops position relative to the cervical wheel facia (Fig 7.11.4).



Fig 7.11.4

7. Check the UBS to ensure the bridge latch mechanism still slides freely in and out of the square hole in the UBS and adjust UBS angle if required.
8. Fasten the UBS in place by tightening the four hex head bolts using a 9/16" wrench.
9. Verify that the latch tab moves freely by depressing and releasing it by hand and docking and undocking the table. If the latch tab does not fully protract or is rubbing the latch fixture of the patient support, the UBS adjustment must be repeated.

7.12 No-undock Hook Adjustment

The no-undock hook is an interlock which prevents the stretcher from being undocked when the patient support is not in the home position. By hand, move the hook up and down to familiarize yourself with the range of travel.

1. Dock the system, close bridge and advance the patient support roughly 6"-8" into the scanner. While the patient support is advanced and the no-undock hook is in the down position (Fig 7.12 1) against its travel stop, pull the cable firmly downward and clamp it to the no-undock hook by tightening the round head screw with a 5/32" Allen key (Fig 7.12.1).

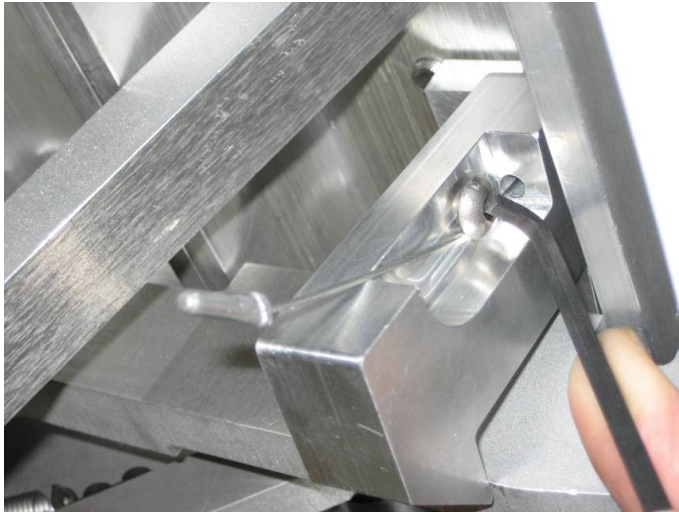


Fig 7.12.1

2. Reach up and depress the travel stop plunger fully by hand and check the no-undock hook to verify it touches its travel stops at both ends of its range of motion. It should be at least 1/4" clear of the undock pedal arm when the patient support is in the scanner, and completely clear of the undock pedal arm when the patient support is at the home position.
3. Check to ensure the hook is not overloaded when the stretcher is in the home position by pulling the patient support in the head-foot direction. There should be a small amount of free play, and the latch fixture should not be pressed against the latch tab. If the no-undock hook is overloaded, the cable should be loosened slightly and these checks repeated.

7.13 Doghouse Interlock Adjustment

The doghouse Interlock is a secondary mechanism that prevents the patient support from being rolled off the stretcher if the dock pedal is held in the depressed position when not docked to the MRI. Adjustment of the doghouse mechanism plunger is always required.

1. Before docking the stretcher, completely depress the doghouse mechanism plunger and take note of the amount of protrusion by viewing it from directly above and using the edge of the Patient Support as a reference.
2. Dock the stretcher to the scanner. As the doghouse advances to latch with the patient support, watch carefully to see if the scanner's white doghouse plunger depresses the doghouse interlock plunger fully and if the doghouse latches freely to the patient support latch fixture. There are two failure conditions that can occur:

- The doghouse plunger can be prematurely depressed. This can overload the doghouse interlock mechanism and will prevent the doghouse from latching to the patient support.
- The doghouse will latch to the patient support but the scanner's white doghouse plunger will not fully depress the doghouse interlock plunger. Take note of the distance required for latching to occur (Fig 7.13.1)

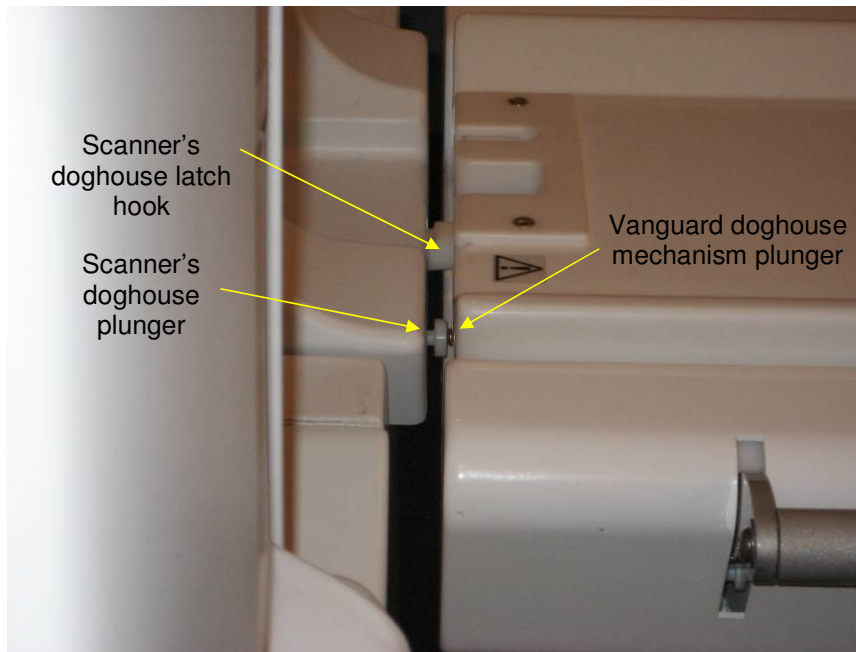


Fig 7.13.1

3. Unlatch the doghouse by hand by holding up the doghouse latch hook while advancing the doghouse into the scanner. Loosen the nut on the brass adjusting screw so that it can be adjusted. By hand or with a non-magnetic screwdriver, screw the brass adjusting screw in or out the amount required to permit latching.
4. Return the doghouse to the home position and ensure that the doghouse plunger is fully depressed but not overloaded. When the doghouse advances to latch to the tabletop, the patient support latch fixture should be latching fully (with a click).
5. Advance the patient support a short distance into the bore. Check to ensure that the patient support has some free play in the head-foot direction ($<1/4"$). If it does not, the doghouse interlock may be overloaded. If there is no movement adjust the doghouse mechanism plunger and try again.
6. Move the table in and out of the scanner. Ensure that the patient support can move in and out of the bore freely and the doghouse mechanism is not rubbing against the

guide rails of the patient support and the scanner's bridge. Apply some downward pressure with one hand on the foot end of the patient support and ensure the doghouse can freely advance the patient support into the scanner without snagging on the doghouse interlock.

7. Once the adjustment is correct, re-tighten the locking nut. *Use a wrench to ensure the nut is snug, while exercising care not to strip the plastic doghouse interlock plunger.*

7.14 Final Verification of Install

Final verification of install is an important series of functional and visual checks which is undertaken to ensure that the installation procedure was executed correctly. If a malfunction is detected during any of the verifications, review and repeat the relevant section of the installation procedure and test again or use **Vanguard Service Manual SMI-0247** to troubleshoot the problem.

1. Replace the drawers. Test them to ensure they open and close smoothly.
2. Verify the function of the bridge latch and locking mechanism by opening and closing the bridge with the patient support in the home position. The tabletop must be prevented from advancing to scan using the MRI's controls when the bridge is open.
3. Verify the proper function of the travel stop plunger mechanism by advancing the patient support into the scanner and depressing the travel stop plunger by hand. It should not show any signs of friction. Verify the travel stop plunger permits the stretcher to undock only when the patient support is at the home position.
4. Verify the no-undock hook is lifted at least 1/4" clear of the undock pedal arm when the patient support is in the home position. The hook must also touch its travel stops at either end of its range of motion.
5. Verify the correct function of the latch tab mechanism by stepping on the dock pedal. The latch tab must retract flush or up to 1/32" above or 1/16" below the top surface of the LBS when the docking pedal is pressed fully and should not rub against any part of the patient support when it is being actuated.
6. Verify the correct function of the doghouse interlock. It must move freely and fully actuate the doghouse mechanism when the doghouse is latched to the patient support without overloading the white plunger on the doghouse.
7. With the stretcher away from the MRI, close the bridge and depress and hold the docking pedal completely. It should not be possible to roll the tabletop off the stretcher with 30lbs (14kg) of force.

8. Ensure the catchment opens fully, does not interfere with any part of the Vanguard and that it closes and stays latched when patient support moves in and out of the scanner.
9. Verify the correct function of the side handles. They must actuate fully between locked positions and the white plastic covers must not leave any gaps larger than 5/16" (8mm)
10. Verify the correct function of the Vanguard docking and undocking. When docked the Vanguard should not come undocked when it is subject to force in any direction. Apply 50lbs of force to the stretcher handle toward the scanner to ensure the Vanguard stays docked.
11. Verify the correct function of the patient support advancing to scan and returning to home. While advancing to scan apply enough resistance to the patient support to cause the doghouse to stop and continue to advance to scan and then return to home position. The patient support should not come unlatched or produce any noise or signs of friction with the scanner bridge or bore.
12. Verify the function of the emergency release by advancing the tabletop to the scan position and then actuating the release to free the table. It should release when the handle is twisted 90 to 110 degrees and the patient support is pulled with minimal force. The patient support should latch securely to the doghouse again after the emergency release handle is released gently.
13. Check that all caster roll and swivel lock brakes work properly.
14. Using the included reference card, install the foam padding to the patient support and ensure that a complete set of pads is present.
15. Insert the fuse which is taped to the side of the battery enclosure to the battery PCB. Verify correct function of all four lights on each side of the stretcher. Ensure there are no non-functioning LEDs
16. Apply wicking Loctite 290 to the two bumper spring housings and doghouse sensor plunger where they screw in to the bumper plate and quickly wipe off the excess from the bumper and bumper spring housings to prevent staining.
17. Review all technical notes, service bulletins and additional procedures related to the Vanguard for GE Signa MRIs and ensure all actions are complete.

7.15 Final Verification Checklist

The following checklist is to be completed as the final set of checks in the verification stage of the installation procedure. A copy of this checklist can be separated and given to customer as verification of installation.

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(Canada)



Installation Verification

The following checks certify the successful installation of the Vanguard on a GE Signa MRI scanner.

- ☐ Lighting system functioning.
- ☐ Hook Height is at least 1/4" higher than docking fixture hook.
- ☐ Eccentric hook height adjuster screw is tightened firmly.
- ☐ Docking force compresses bumper springs, resists undocking attempts.
- ☐ Both docking hook adjustment screws are tight.

- ☐ Bumper is vertically and horizontally aligned to docking fixture.
- ☐ Stretcher and scanner guide rails are colinear.
- ☐ Bumper mounting screws are tight.
- ☐ Bottom left/right fine adjustment bumper screws are tight.
- ☐ Rear left/right fine adjustment bumper screws are tight.

- ☐ 1" gap from scanner bridge to edge of LBS.
- ☐ Latch tab is flush when system is docked, cable clamp is screwed tight.
- ☐ Latch tab moves freely in patient support latch fixture.
- ☐ Patient support slides into bore smoothly with no contact.

- ☐ LBS coarse height adjustment screws are tight.
- ☐ UBS coarse height adjustment screws are tight.
- ☐ Bridge guide block is centered with LBS opening.
- ☐ LBS forward/backward adjustment screws are tight.
- ☐ UBS forward/backward adjustment screws are tight.

- ☐ Doghouse mechanism retains patient support when latch tab is retracted.
- ☐ Doghouse mechanism plunger nut is tight.
- ☐ Doghouse stopper is more forward than travel stop plunger.
- ☐ Doghouse hook has play with patient support latch fixture.
- ☐ Patient Support remains latched to doghouse if motion is halted.

- ☐ No-undock hook is not contacting pedal arm in down position.
- ☐ No-undock hook cable is screwed tightly to no undock hook.
- ☐ No-undock hook is clear of pedal – touching hard stop when actuated.

- ☐ Caster mount fine adjustment nuts are very tight.
- ☐ Casters are firmly mounted, undamaged and rolling/swiveling smoothly.
- ☐ Caster covers are tight and flush with frame.
- ☐ Emergency release works with <110 degree turn and light pull force.
- ☐ Loctite 290 applied to spring bumpers and sensor actuator plunger.
- ☐ System is cleaned, free of fingerprints or debris.

- ☐ 8ch bilat SNR is _____ (Acceptance = >403)
- ☐ 4ch bilat SNR is _____ (Acceptance = >333)
- ☐ 2ch left SNR is _____ (Acceptance = >280)
- ☐ 2ch right SNR is _____ (Acceptance = >280)
- ☐ 8ch SNR is entered in customer user manual.

- ☐ Padding kit is installed correctly.
- ☐ All parts and accessories are included and shortages reported immediately.
- ☐ Calling card is filled out and left on console desk.

- *The following two checks apply only to scanners with 12.0 software**
- ☐ Vibrant is enabled.
- ☐ Correct coil configs are modified.

The Vanguard is now successfully installed. It may only be used after Sentinelle Applications training is complete.

Customer: _____

Serial #: _____

Date: _____

Installed by: _____

Signed: _____

Accepted by: _____

Signed: _____

7.16 RF Coil Testing

1. Install the compression frames, medial array and lateral arrays, consulting the included user manual.
2. Test the RF system by executing Site Installation SNR Testing Procedure SMI-0217.
3. Perform all other imaging tests provided with the service manual for this product.

7.17 Departure

1. Ensure work area is clean and no tools or garbage are left behind.
2. Replace Signa MRI stretcher.
3. Complete a Sentinelle calling card and leave it in a prominent place, preferably at the console.
4. If possible, inform staff member on site of completion before leaving.
5. Ensure doors (especially scanner room doors) are locked and lights are off in accordance with the site's requirements.